Robotics Lab - 221 LIA 001

Assignment 2
Due:October 27, 2026, 11.00 pm IST
Submission via classroom50

- **Question 1:**
  On August 27 2023, the Indian Space Research Organisation (ISRO) released a graph of the temperature variation between the moons surface and a point around 8 cm below as measured by an instrument named ChaSTE on board the lander module of the Chandrayaan-3 mission [1][2]. ChaSTE is a temperature probe that can be driven into the moons surface using a motor to a depth of up to 10 cm. The temperature probe has 10 sensors.

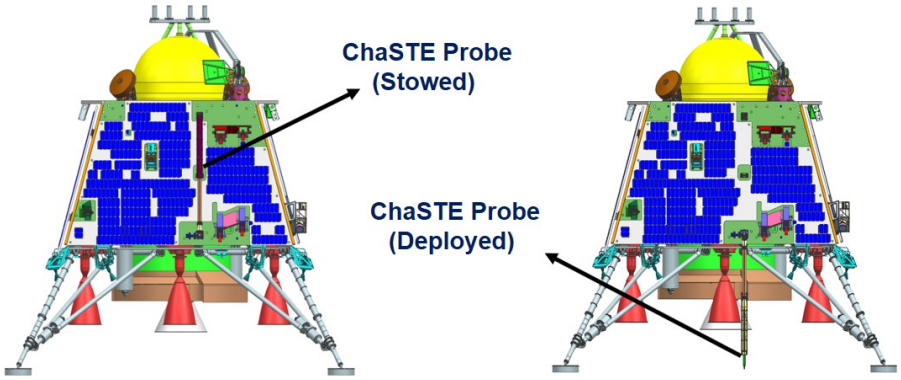


Figure 1: Chandra's Surface Thermophysical Experiment (ChaSTE) probe [1]

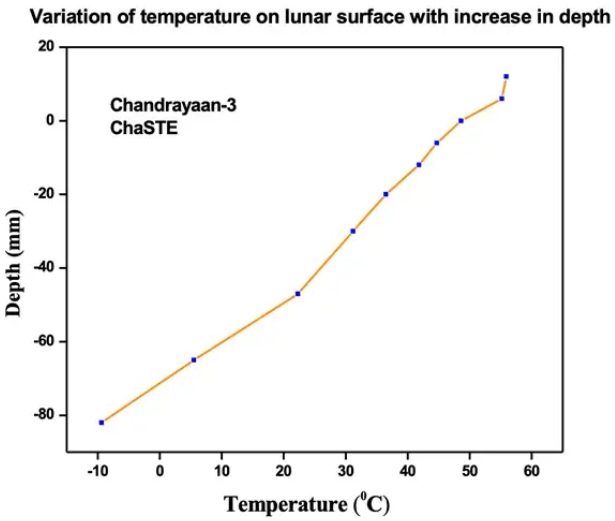


Figure 2: Variation of temperature on lunar surface with increase in depth [2]

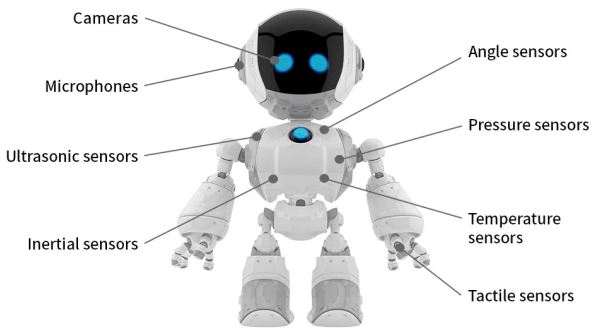
The question 1 of assignment is inspired from the ChaSTE probe on-board the Vikram lander. Assuming that the above system is ROS 1 based, we simulate the temperature acquisition using ROS publisher-subscriber model involving two nodes 'temperature_sensor' and 'control_unit_01'. The 'temperature_sensor' node publishes temperature updates acquired by 'probe_01' on topic 'temperature_update' at rate of 1.0 Hz. The 'control_unit_01' node subscribes to the 'temperature_update' topic.

1. List the currently running nodes
2. List the currently running topics
3. Run the rqt_graph tool and paste the ros graph'
4. Use the built-in screen recorder, and record a **35 second** of the terminals showing the nodes running. Save the file in *webm* format as <your name>_assign2_2.webm as commit in the assignment submission.
   (Refer this link on how to save videos using Ubuntu 20.04's built-in screen recorder)

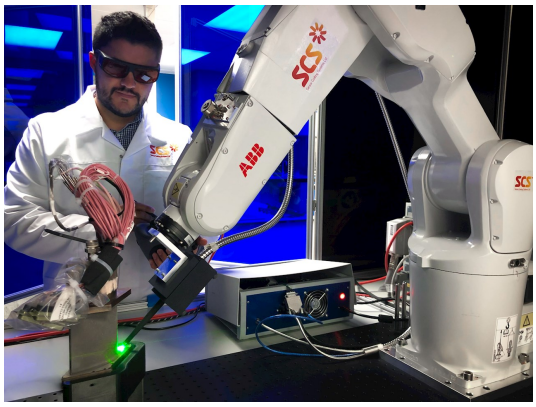
Hints :

1. Use suitable ROS message types from *sensor_msg* package to simulate the system
2. The random temperature values can be generated using methods from *random* Python module like *random.randrange()* or *random.random()*
3. Refer the documentation available at ROS Wiki

- More examples involving robots in sensing applications

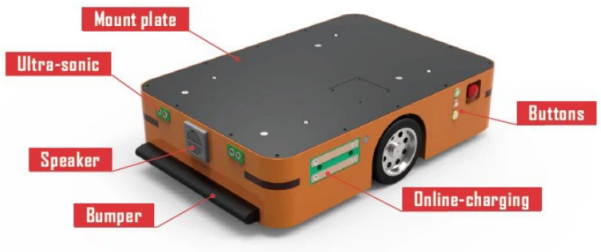


(a) Latest service robots are equipped with dozens of sensors for purposes like spatial awareness and motion detection [3]



(b) ABB IRB 1200 robot in non-invasive extreme temperature sensing application [4]

- **Question 2:**
  In industrial warehouses and automated shop floors, Automated Guided Vehicles (AGVs) rely on proximity sensors to avoid collisions with workers and other equipment. If an obstacle is detected within a critical threshold distance, the robot controller must immediately override the drive command and halt the vehicle.



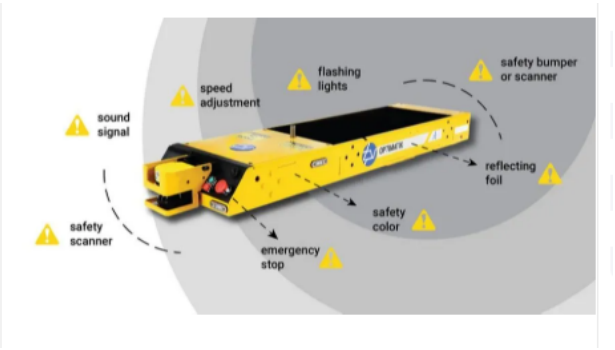


Figure 4: Sensors on Automated Guided Vehicles

Assuming that the mobile platform is ROS 1 based, we simulate an emergency obstacle detection and stopping mechanism using two nodes: 'obstacle_detector' and 'motion_controller'.

The 'obstacle_detector' node simulates an ultrasonic range finder mounted in front of the robot and publishes distance measurements on the topic 'front_sonar' at a rate of 5.0 Hz. Make use of proper message type for this purpose from ROS 1 *sensor_msgs*.

The 'motion_controller' node subscribes to 'front_sonar'. When the distance falls below a safe margin of 0.30 m, it triggers an emergency stop by halting motor commands and displaying a warning message on the terminal.

1. List the currently running nodes
2. List the currently running topics
3. Run the rqt_graph tool and paste the ros graph'
4. Use the terminal command `rostopic pub` to manually publish a distance value of 0.20 m on the 'front_sonar' topic. Note down the output message displayed on the controller terminal.
5. What message type did you select from *sensor_msgs* package and what are its essential fields?

Hints :

1. Refer the documentation available at ROS Wiki for *sensor_msgs*
2. The sensor distance values can be generated using *random* module or decreased steadily in a loop to simulate an approaching wall

- **Question 4:**
In robotic sensing tasks such as drilling or temperature profiling, sensors often have initial bias or drift that must be zeroed before starting an experiment. Unlike continuous data streaming using topics, tare or calibration is an on-demand operation performed only when commanded. In ROS, such two-way request-response interactions are handled using ROS Services.

  In this question, we realize an on-demand calibration setup using a service server node 'sensor_calibrator' and a client node 'mission_control_client'. The 'sensor_calibrator' node advertises a service named 'zero_sensor' using the standard ROS service type *std_srvs/Trigger*. When called by the client, the server node calculates an offset value, resets the sensor reference, and returns a response confirming success along with a status message.

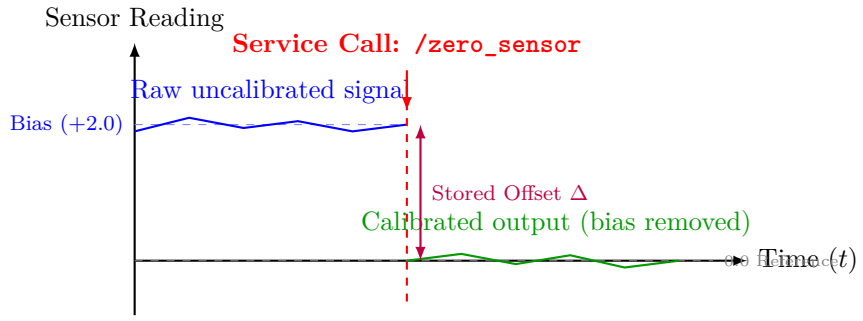


Figure 5: Sensor telemetry before and after on-demand zero calibration

1. List all currently available ROS services using the ROS command-line tool
2. Call the 'zero_sensor' service directly from the terminal without running the client Python script. Paste the terminal output and response received.
3. Run both the server and client Python scripts and commit the terminal outputs
4. Explain the fundamental difference between ROS topics and ROS services. Mention one real practical scenario where a service is preferred over a topic.

Hints :

1. Use *Trigger* service type from the *std_srvs* package
2. In Python, refer to *rospy.Service* for creating the server and *rospy.ServiceProxy* for writing the client node
3. Refer the documentation available at ROS Wiki on ROS Services and rospy tutorials

References

[1] @online Chandra's Surface Thermophysical Experiment (ChaSTE), https://www.prl.res.in/~pids/ChaSTE.html# Online; accessed 24-October-2023

[2] @online The first observations from the ChaSTE payload onboard Vikram Lander, https://www.isro.gov.in/Ch3_first_observation_ChaSTE_Vikram_Lander.html#:~:text=ChaSTE%20(Chandra's%20Surface%20Thermophysical%20Experiment,10%20cm%20beneath%20the%20surface. Online; accessed 24-October-2023

[3] @online What Role Do High-Precision Sensors Play in Creating Natural Movement in Robots?, https://www.tdk.com/en/featured_stories/entry_037.html Online; accessed 24-October-2023

[4] @online It is rocket science: ABB robot helps test extreme high temperature systems, https://new.abb.com/news/detail/61133/it-is-rocket-science Online; accessed 24-October-2023

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October 19, 2026